

Discovery Center

Installation and Operations Manual

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Synthetic demonstration document · not real product documentation.

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1 Introduction

This manual describes the installation, configuration and routine operation of the Discovery Center management platform. It is intended for deployment engineers and operations staff who are already familiar with radio access network terminology.

The platform manages a mixed estate of gNodeB and eNodeB elements and presents a single operational view across both. See Figure 3 for an overview of the deployment topology.

Conventions used throughout this document are listed in Appendix A. Parameter names appear in fixed width, for example `airscale_rnc`, and interface names such as X2 appear unmodified.

2 System Architecture

The platform is built from four cooperating services: the collector, the correlation engine, the operational data store and the presentation layer. Each service may be scaled independently.

Traffic between the collector and the data store is compressed and checkpointed, so a restart of either component does not lose counters that have already been acknowledged.

2.1 Network Elements

Supported elements include gNodeB, eNodeB and the SGW. Element records are keyed on the distinguished name reported by the element itself rather than on the address it happens to be reachable at.

Where an operator shares a radio network between core operators using MOCN, each participating operator is modelled separately so that counters are attributed correctly.

2.2 Interfaces

Northbound interfaces expose the operational data store over a versioned interface. Southbound interfaces collect from elements on a polling cycle whose period is configurable per element class.

Signal quality measurements such as RSRP are collected on the standard cycle and retained at full resolution for thirty days.

3 Installation

Installation is performed in two stages. The first stage provisions the operating system prerequisites; the second installs the platform packages and performs first-time initialisation.

Before starting, ensure that the license file is there before you begin, since initialisation cannot complete without it.

3.1 Hardware Prerequisites

A minimum of eight physical cores and 32 GB of memory is required for a production installation. Laboratory installations may be run on four cores, with reduced retention.

Storage must be provisioned as a single volume. Splitting the data store across volumes is not supported and will fail validation at startup.

3.2 Software Prerequisites

The supported operating system baselines are listed in Table 2. Applying vendor security patches is supported and expected; upgrading the kernel major version is not.

The installer verifies each prerequisite before making any change, and reports every unmet condition in a single pass rather than stopping at the first failure.

4 Configuration

Configuration is held in a single versioned document. Changes are staged, validated and then committed, so a rejected change never leaves the system in a partially applied state.

Refer to Section 9.4 for the full parameter reference.

4.1 Initial Setup

On first start the platform creates an empty inventory and waits for elements to be registered. Registration may be performed element by element or in bulk from a prepared file.

A diagnostic report is produced at the end of initial setup. When the parameter was retrieved from the OSS database, the report records both the requested value and the value actually applied.

4.2 Access Control and Identity Verification

Every operator account is verified against the configured identity provider before any session is established. Credentials are never held by the platform itself; only the assertion returned by the provider is retained, and only for the lifetime of the session.

Authentication failures are recorded with the originating address and the reason reported by the provider. Repeated failures from one address cause that address to be held off for an increasing interval.

Where an operator requires two-factor verification, the second factor is requested by the identity provider rather than by the platform, so no additional configuration is needed here.

5 Operations

Routine operation is largely unattended. The tasks described in this section are the ones that require an operator decision.

5.1 Monitoring and Alarms

Alarms are raised against the element that reported the underlying condition, not against the collector that observed it. An alarm clears only when the reporting element confirms the condition has ended.

Alarm severity is taken from the element where the element supplies one. Where it does not, severity is derived from the alarm class as shown in Table 4.

5.2 Performance Counters

Counters are collected on the polling cycle and aggregated at one minute, one hour and one day. Aggregates are computed from retained samples, so a late-arriving sample corrects every aggregate that contains it.

6 Troubleshooting

This section lists the conditions most often reported to support, and the checks that resolve them.

6.1 Collector Cannot Reach an Element

Confirm first that the element is reachable from the collector host and not merely from the operator workstation. The collector does not share the workstation's routing.

If the element is reachable but registration still fails, the distinguished name reported by the element has probably changed.

6.2 OSS Database Errors

Errors reported by the operational data store are recorded with the originating query and the connection identifier. Both are needed when raising a support case.

A sustained rise in rejected writes usually indicates that retention has not been applied, rather than a fault in the data store itself.

Appendix A: Glossary

gNodeB · 5G base station. eNodeB · LTE base station. MOCN · Multi Operator Core Network. RSRP · Reference Signal Received Power. SGW · Serving Gateway. X2 · interface between base stations.

airscale_rnc · the radio network controller parameter group.